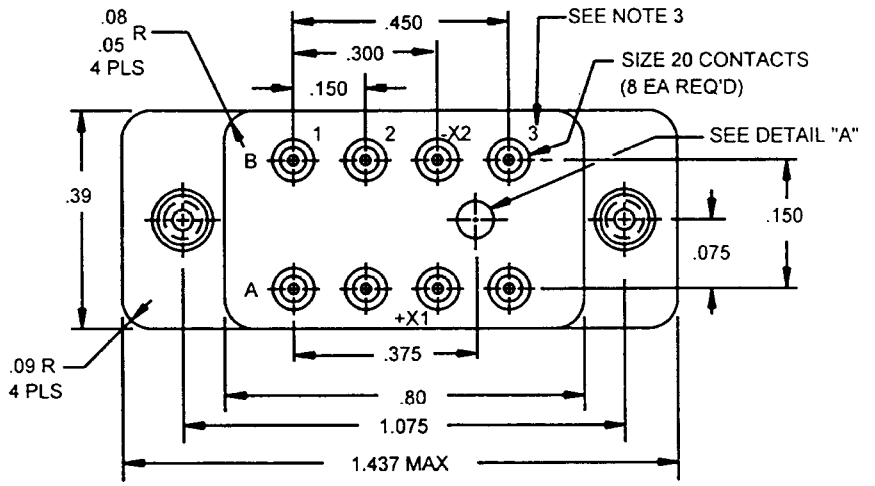
INCH-POUND
MIL-PRF-12883/45C
28 June 1996
SUPERSEDING
MIL-S-12883/45B
20 December 1989

PERFORMANCE SPECIFICATION SHEET

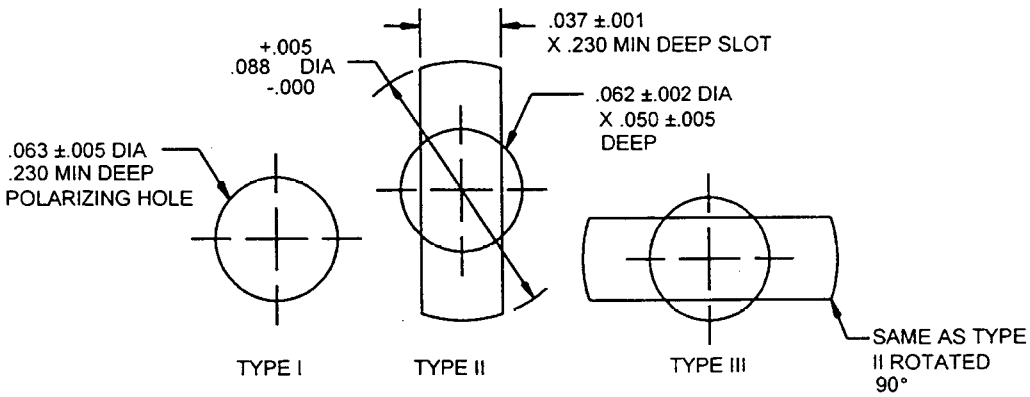
# SOCKETS, PLUG-IN ELECTRONIC COMPONENTS, FOR RELAYS, FOR 2-POLE, 5 AMPERES, (MIL-R-6106 AND MIL-R-83536)

This specification is approved for use by all Departments and agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-PRF-12883.




MATING FACE

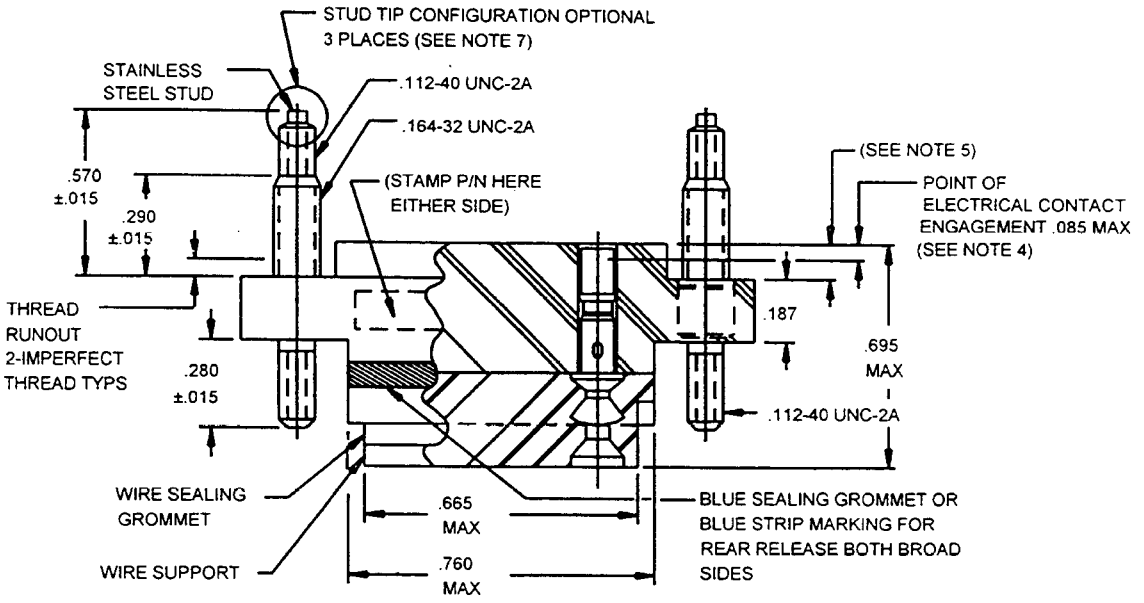



DETAIL "A"

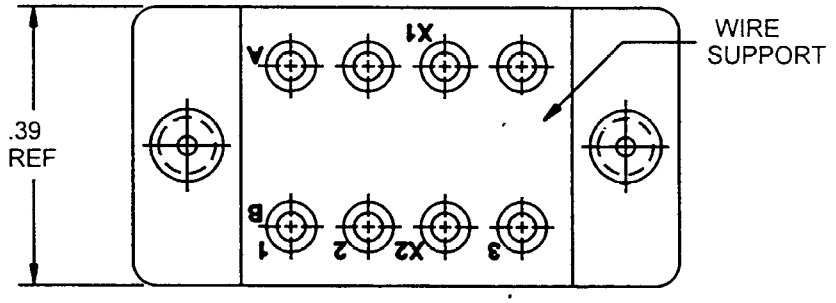
FIGURE 1. Socket configuration (-01, -02, -03).


DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

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SIDE VIEW

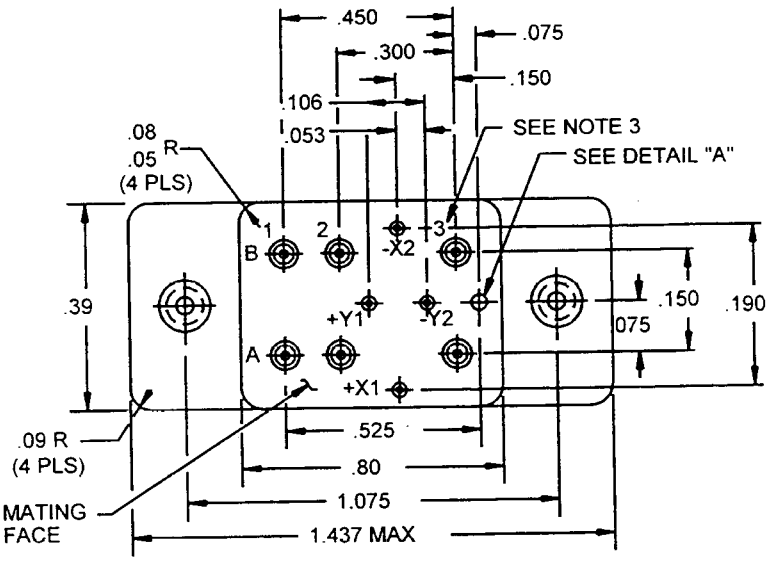



BOTTOM VIEW

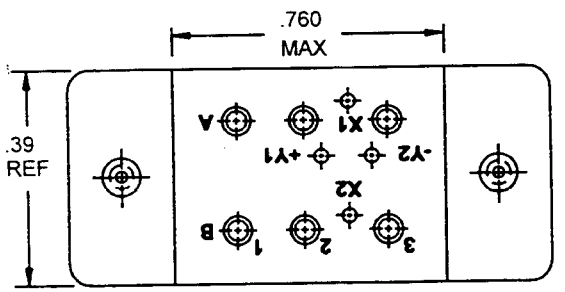
FIGURE 1. Socket configuration (-01, -02, and -03) - Continued.



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MATING FACE




BOTTOM VIEW

| Inches | mm |
|---|---|
| .001 | 0.03 |
| .002 | 0.05 |
| .005 | 0.13 |
| .01 | 0.25 |
| .015 | 0.38 |
| .037 | 0.94 |
| .05 | 1.3 |
| .050 | 1.27 |
| .053 | 1.35 |
| .062 | 1.57 |
| .063 | 1.60 |
| .075 | 1.90 |
| .08 | 2.0 |
| .085 | 2.16 |
| .088 | 2.24 |
| .09 | 2.3 |
| .093 | 2.36 |
| .106 | 2.69 |
| .112 | 2.84 |
| .113 | 2.87 |
| .150 | 3.81 |
| .164 | 4.17 |
| .187 | 4.75 |
| .190 | 4.83 |
| .230 | 5.84 |
| .280 | 7.11 |
| .290 | 7.37 |
| .300 | 7.62 |
| .375 | 9.52 |
| .39 | 9.98 |
| .450 | 11.43 |
| .570 | 14.48 |
| .665 | 16.89 |
| .695 | 17.65 |
| .760 | 19.30 |
| .80 | 20.3 |
| 1.075 | 27.31 |
| 1.437 | 36.50 |

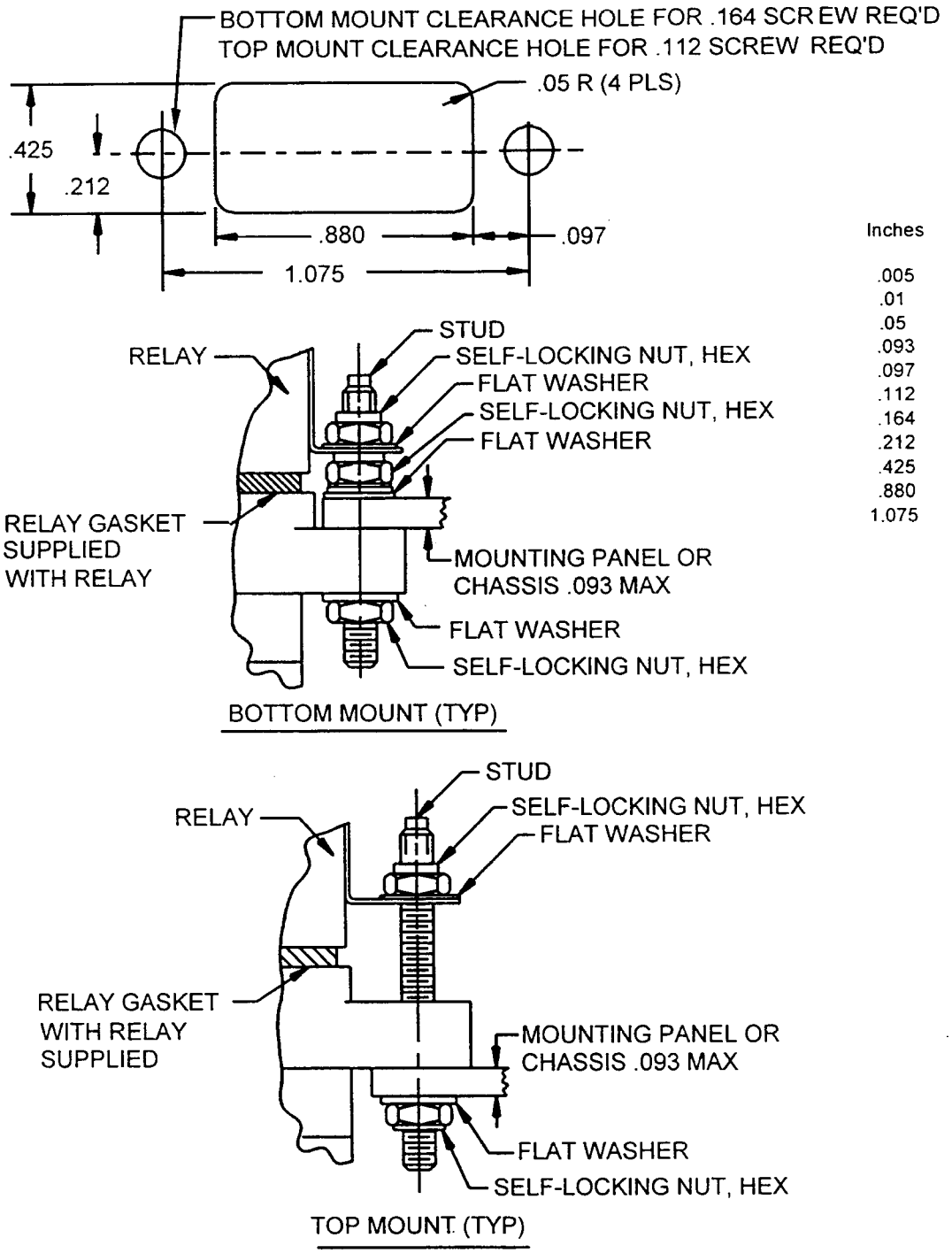
NOTES

1. Dimensions are in inches.
2. Unless otherwise specified tolerances are ±.005 inch (0.13 mm) for three place decimals and ±.01 inch (0.25 mm) for two place decimals.
3. Marking shall be characters which are molded .035 inch (0.89 mm) minimum. Ink marking optional (see MIL-STD-1285).
4. Point of electrical contact engagement, from mating face of socket insulator to the socket contact.
5. -01 through -03 .093 inch (2.36 mm); -04 through -06 .113 inch (2.87 mm).
6. Metric equivalents are given for general information only.
7. The configuration of the stud tip shall be optional provided it meets all the requirements of this specification.

FIGURE 2. Socket configuration (-04, -05, and -06).



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| Inches | mm |
|---|---|
| .005 | 0.13 |
| .01 | 0.25 |
| .05 | 1.3 |
| .093 | 2.36 |
| .097 | 2.46 |
| .112 | 2.84 |
| .164 | 4.17 |
| .212 | 5.38 |
| .425 | 10.80 |
| .880 | 22.35 |
| 1.075 | 27.31 |

NOTES
1. Dimensions are in inches
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerances are ±.005 (0.13 mm) for three place decimals and ±.01 (0.25 mm) for two place decimals.

FIGURE 3. Mounting (hardware and panel cutout).



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REQUIREMENTS:

Design and construction: See figures 1, 2, 3, and table I.

Reference material: (see 3.1.1 of MIL-PRF-12883)

Insulator and wire support: Diallyl phthalate, in accordance with MIL-M-14, type SDG-F or any glass filled thermoplastic material in accordance with MIL-P-46179, or an approved equivalent material manufactured to a commonly accepted industry standard.

Color: Material color shall be optional providing that the color provides a contrasting background for the blue sealing grommet or blue color bands indicating rear release contacts.

Grommet: Silicon rubber in accordance with ZZ-R-765.

Mounting hardware: Corrosion resistant steel or steel with cadmium/chromate finish.

Electrical:

Test gauge: A mating relay in accordance with table I shall be used as a test gauge or mating connector.

Insulation resistance: 1000 megohms minimum.

Dielectric withstanding voltage:

Sea level: Test voltage, 1000 V rms.

High altitude (80, 000 feet (24,384 meters)): Test voltage, 500 V rms.

Environmental:

Temperature range: Operating temperature range -70°C to +125°C.

Wire sealing: A resilient socket grommet is permanently bonded to the wire entry face of the socket seas to provide moisture sealing capabilities, over the following range of finished wire;

| Contact | Wire diameter |
|---|---|
| M39029/101-552 | .030 min - .060 max (0.76 min - 1.52 max mm) |
| M39029/101-553 | .040 min - .083 max (1.02 min - 2.11 max mm) |

Mechanical:

Contacts: In accordance with MIL-C-39029/101 (see table I) , or approved equivalent contacts manufactured to a commonly accepted industry standard. Contacts shall be crimp removable type, rear release and accept relay pins size 20 (.039 through 0.41 inch (0.99 through 1.04 mm)) and size 22 (.030 through .032 inch (0.76 through 0.81 mm)).



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TABLE I. Dash numbers and characteristics.

| Dash number | Socket configuration | Contact size mating end wire barrel | | Number of contacts | Contact designation M39029/101 | Mating relay MIL-R-83536 1/ | Polarization type |
|---|---|---|---|---|---|---|---|
| 01 | -01 | 20 | 20 | 8 | -553 | /1-021, /1-024, /2-021, /2-024 | I |
| 02 | -02 | 20 | 20 | 8 | -553 | | II |
| 03 | -03 | 20 | 20 | 8 | -553 | | III |
| 04 | -04 | 20 | 20 | 6 | -553 | MIL-R-6106 /38-002 | None required |
| | | 22 | 22 | 4 | -552 | | |
| 05 | -05 | 20 | 20 | 6 | -553 | | II |
| | | 22 | 22 | 4 | -552 | | |
| 06 | -06 | 20 | 20 | 6 | -553 | | III |
| | | 22 | 22 | 4 | -552 | | |

1/ Reference MIL-R-83536 for supersession data on MIL-R-6106 relays.

Insertion and withdrawal force: The insertion and withdrawal forces shall be as specified in table II.

TABLE II. Insertion and withdrawal forces.

| M12883/45 | -01 through -03 | -04 through -06 |
|---|---|---|
| Initial insertion force (max) | 9 lbf (40.0 newton) | 10.0 lbf (44.5 newton) |
| Withdrawal force (min) | .35 lbf (1.6 newton) | .44 lbf (2.0 newton) |
| Insertion force (max), after 10 insertions and withdrawals, before vibration | 11.0 lbf (48.9 newton) | 11.7 lbf (52.0 newton) |
| Withdrawal force (min) | .30 lbf (1.3 newton) | .38 lbf (1.7 newton) |
| Insertion force (max), after vibration | 11 lbf (48.9 newton) | 11.7 lbf (52.0) |
| Withdrawal force (min) | .30 lbf (1.3 newton) | .38 lbf (1.7 newton) |

Durability: Shall be in accordance with MIL-PRF-12883.

Vibration (sinusoidal): In accordance with MIL-STD-202 method 204, test condition G, or an equivalent industry standard.

a. Except that the frequency range shall be varied logarithmically between the limits of 10 and 3000 HZ.

b. Except that the procedure of method 201 of MIL-STD-202, or an equivalent industry standard, may be applied during the 10-55 Hz band of the vibration frequency range.



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Vibration, (random): In accordance with MIL-STD-1344, method 2005, test condition V, letter G, or an equivalent industry standard, with a duration of 15 minutes. The mating relay shall be used as a test gage.

Shock (mechanical): In accordance with MIL-STD-202 method 213, condition C, or an equivalent industrystandard, except peak value shall be 200 g's.

Torque: Relay sockets and hardware shall be subjected to testing torque as specified in table III. Sockets shall be installed in mounting panel when test torque is applied. No physical damage will be permitted. Torque shall be maintained for a reasonable period of time to insure stud, socket, and associated hardware have not been damaged.

TABLE III. Torque requirements (installed in panel conditions).

| Thread size | Torque | | | |
|---|---|---|---|---|
| | Testing | | Installation | |
| | Inch-pounds | Newton-meters | Inch-pounds | Newton-meters |
| .112-40 | 8 + 1 - 0 | 0.90 + .11 -0 | 4 ± 1 | 0.45 ± .11 |
| .164-32 | 20 + 1 - 0 | 2.26 + .11 - 0 | 15 ± 1 | 1.70 ± .11 |

Socket, contact tools: See table IV, approved equivalent industry standard tools may also be used whereappropriate.

TABLE IV. Contact tools.

| Nomenclature | Part or Identifying Number (PIN) |
|---|---|
| Crimp tool | M22520/7-01 |
| Positioner | M22520/7-11 (for size 22)<br>M22520/7-12 (for size 20) |
| Insertion and removal tool | M81969/16-01 (for size 20)<br>M81969/16-04 or<br>M81969/14-01 (for size 22) |

Weight: The socket, hardware, and electrical contacts shall have a maximum weight of .033 pounds (15 grams).

Mounting hardware: The mounting hardware shall allow mounting the socket above or below the panel or chassis (see figure 3), and shall allow mounting and securing the relay to the socket without disturbing the mounted socket or access to the wiring side of the socket. The hardware shall provide the nominal spacing between socket surface and relay mounting flange, regardless of mounting configuration.

Mounting hardware shall be supplied with the relay socket and shall consist of the following:

Self locking nuts .112-40 (4 each) (.206 max dia x .176 max height).

Flat washers .112 (4 each) (.224 max O.D. x .021 max thickness).

Self locking nuts .164-32 (2 each) (.290 max dia x .190 max height).

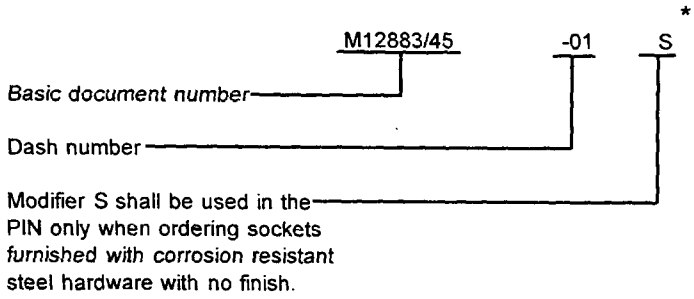
Flat washers .164 (3 each) (.290 max O.D. x .019 max thickness).



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PIN: The PIN shall be marked on the socket body as shown in the example (see figure 1). The PIN shall consist of the basic number of this specification sheet and the dash number from table I.

Example




* For future acquisition of these sockets as of the effective date of revision B, 20 December 1989, parts identified with an "S" modifier shall be corrosion resisting steel (CRS), and parts without an "S" modifier shall be cadmium chromate finish. No mixing of hardware types shall be permitted.

Ordering data: Sockets without contacts may be ordered when so indicated in the ordering data (see MIL-PRF-12883). This applies only to original equipment manufacturers (OEM's) and subcontractors. All direct shipments to the government shall include all applicable contacts and mounting hardware. The PIN to be marked on the socket shall be as shown in the PIN example (see figure 1 and table I).

Revision letters are not used to denote changes due to the extensiveness of the changes.

TABLE VI. Supersession and cross reference.

| Active Government PIN | Superseded manufacturers part number | |
|---|---|---|
| | CAGE 58982 | CAGE 99699 |
| M12883/45-01 | RSE120159 | SE205-1010 |
| M12883/45-02 | RSE120161 | TBD |
| M12883/45-03 | RSE120163 | TBD |
| M12883/45-04 | RSE120165 | TBD |
| M12883/45-05 | RSE120167 | TBD |
| M12883/45-06 | RSE120169 | TBD |

CONCLUDING MATERIAL

Custodian:
ARMY - CR
NAVY - EC
Air Force - 85

Review activity:
Army - AR, AT, AV, ME
Navy - AS, MC, OS, SH
Air Force - 99

Preparing activity:
DLA - ES

(Project 5935-4015-08)

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